

Cost-Optimized Menu Recommendation System using Machine Learning

by

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ABSTRACT

Fluctuations in agricultural commodity prices, driven by extreme weather, macroeconomic shifts, and seasonal festivals, frequently subject consumers and food vendors to unexpected market shocks. Currently, there is a lack of accessible tools capable of translating these complex price predictions into actionable meal planning. To address this gap, this research develops a "Cost-Optimized Menu Recommendation System" utilizing Machine Learning. The primary objective is to forecast the prices of 13 key ingredients 7 days in advance and subsequently rank 20 popular Thai menus based on their predicted total computational costs.

The methodology integrates a comprehensive 5-year time-series dataset (2021–2025) comprising daily agricultural prices, localized meteorological data from four strategic provinces, diesel prices, and a custom calendar of significant events. Through rigorous data preprocessing, including temporal feature engineering and time-series imputation (Forward/Backward Fill), the system effectively resolves structural missingness in market reporting. Exploratory Data Analysis (EDA) yielded critical insights, actively challenging the assumption of immediate, linear weather impacts. Instead, the data revealed that meteorological effects are highly non-linear and subject to delayed time-lags, whereas diesel prices act as a direct, systemic anchor for livestock costs.

To handle the identified severe spatial multicollinearity and non-linear threshold effects, a Random Forest Regressor is established as the robust baseline model, which will be systematically compared against Recurrent Neural Network (RNN) architectures to capture deep sequential dependencies. Ultimately, the most significant achievement of this study is shifting the analytical paradigm from isolated ingredient price tracking to practical "Menu Cost" optimization. By dynamically calculating and ranking recipe costs, the system provides a highly utilitarian tool that empowers users to proactively avoid overpriced ingredients and optimize their weekly food budgets.

Keywords: Machine Learning, Agricultural Price Forecasting, Cost-Optimized Menu, Exploratory Data Analysis, Feature Engineering

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Price volatility of fresh foods (e.g., pork, vegetables, and eggs) significantly impacts the cost of living for consumers and small business operators. These fluctuations are driven by external factors such as seasonal weather conditions, transportation costs, and demand spikes during major cultural festivals. Currently, consumers lack accessible forecasting tools, exposing them to unexpected "market shocks" where ingredient prices surge. This results in inefficient budget planning due to the absence of a unified system that translates complex external market factors into actionable, cost-optimized meal recommendations.

1.2 Research Questions

This study aims to address the following research questions:

1. How can machine learning analyze external factors to forecast ingredient prices and recommend cost-optimized meals?
2. Which external factors (weather, fuel, festivals) have the most significant impact on the price volatility of specific ingredients?
3. What is the optimal forecasting horizon that balances model accuracy with practical budget planning?

1.3 Objectives of the Study

1. To develop an ML-based price forecasting and cost-optimized menu recommendation system to mitigate the impact of market volatility.
2. To quantify the influence of exogenous factors on specific fresh food prices.
3. To determine the optimal forecasting horizon that maximizes both machine learning accuracy and practical usability.

CHAPTER 2

LITERATURE REVIEW

Recent advancements in agricultural commodity price forecasting have increasingly shifted from traditional linear models, such as ARIMA, toward Machine Learning (ML) approaches to better capture complex market nonlinearities and temporal shifts (Chen, 2021; Zhao, 2024). While state-of-the-art deep learning architectures like RNNs and LSTMs excel in managing long-term sequential dependencies, ensemble methods like Random Forest remain highly effective in processing multivariate tabular data, including critical exogenous drivers such as meteorological factors, seasonal calendar events, and embedded logistics costs like diesel prices (Le Ngoc, 2023; Watson Benjamin, 2024). Despite these technical advancements, a significant research gap persists: existing literature predominantly focuses on theoretical point forecasting and often overlooks systematic missing data handling in volatile markets. Consequently, there is a lack of actionable, consumer-focused utility, which this study directly addresses by shifting the paradigm toward a practical, cost-optimized menu recommendation system. Most existing research on price forecasting primarily focuses on government policy and agricultural stakeholders, creating an application gap in that predicted prices are not effectively translated into "cost-efficient meal recommendations" for consumers and small food businesses. Furthermore, current models tend to emphasize factors such as weather conditions and fuel prices, but there remains a variable gap due to the limited incorporation of "cultural festivals" as key drivers of sudden price spikes. In addition, a strategic gap persists, as there is insufficient research identifying the "optimal forecasting horizon" (e.g., one week) that can balance the predictive accuracy of AI models with the practical needs of consumers in planning their weekly budgets.

CHAPTER 3

METHODOLOGY

3.1 Data Collection & Understanding

To construct a robust multivariate time-series dataset, daily records spanning a five-year period (Jan 2021 – Dec 2025, totaling 1,827 rows) were aggregated from primary sources. The dataset integrates average retail prices for 13 key ingredients from Thailand's Department of Internal Trade (DIT), strategic regional meteorological data (temperature, rainfall, humidity, soil moisture) via the Open-Meteo API, daily high-speed diesel prices from Bangchak Corporation representing logistics costs, and a custom calendar flagging significant Thai cultural festivals. Given that the full variables comprise continuous numerical data and binary features with inherent structural missingness during non-business days, a detailed breakdown of all features and their data types can be referenced in the Data Dictionary (Appendix D).

3.2 Exploratory Data Analysis (EDA) Findings

Before preprocessing the data, an extensive Exploratory Data Analysis (EDA) was conducted to understand the underlying market dynamics, distributions, and structural relationships. A comprehensive summary of all EDA findings is also detailed in **Appendix F, Table F.1**. The insights gained directly dictated the subsequent feature engineering strategies:

- **Spatial Multicollinearity in Weather Data:** The Pearson Correlation Matrix (see **Appendix E, Figure E.1**) revealed severe multicollinearity ($r > 0.8$) among meteorological variables across the four provinces.
 - **Interpretation & Action:** This indicates that the weather across these adjacent regions shares nearly identical macro-climatic cycles. Consequently, this justified the decision to aggregate the 16 provincial weather features into 4 macro-climate variables to reduce model noise.
- **Macroeconomic Drivers (Diesel Correlation):** Additionally, the heatmap (see **Appendix E, Figure E.1**) highlighted a notable positive correlation between daily high-speed diesel prices and livestock prices (e.g., chicken, pork, eggs).
 - **Interpretation & Action:** This confirms that fuel acts as an embedded logistics cost that heavily anchors fresh food pricing, justifying its inclusion as a primary continuous input feature.
- **Data Distribution, Skewness, and Outliers:** Histogram and boxplot analyses (see **Appendix E, Figures E.2 and E.3**) demonstrated that while staple proteins exhibit normal distributions, highly volatile ingredients like Chili (Skew: 2.21) and Lime (Skew: 1.35) have long right-tails with extreme upper outliers.
 - **Interpretation & Action:** Longitudinal time-series plots (see **Appendix E, Figure E.4**) confirmed these outliers are not random errors but episodic "market shocks." This crucial finding proved that weather and fuel alone cannot predict extreme price surges, directly leading to the engineering of the Important_Days binary flag to help the model anticipate festival-driven demand spikes.

3.3 Data Preprocessing & Feature Engineering

To ensure the dataset was optimized for machine learning algorithms, rigorous data cleaning and feature engineering were performed. The specific techniques applied, the rationale behind them, and the resulting outcomes are systematically detailed in Table 3.1 below.

Table 3.1. Summary of Data Preprocessing and Feature Engineering Pipeline

Techniques Applied	Rationale / Problem Addressed	Outcome / Result
Missing Value Imputation: Applied Forward Fill (ffill) to agricultural prices during weekends and public holidays	Market prices are only published on business days. Standard interpolation causes data leakage by peeking into future prices.	A continuous, daily time-series dataset with zero missing values, perfectly preserving chronological integrity without data leakage.
Weather Feature Aggregation: Averaged 4 climatic metrics (Temperature, Rainfall, Humidity, Soil Moisture) across the 4 provinces.	Exploratory Data Analysis (EDA) revealed severe spatial multicollinearity ($r > 0.8$), meaning the weather across these adjacent provinces behaves almost identically.	Reduced 16 highly redundant provincial weather columns into 4 robust "macro-climate" features, significantly decreasing model complexity and noise.
Temporal Decomposition & Standardization: Converted Buddhist Era (BE) to Common Era (CE) and extracted Day, Month, Year, and DayOfWeek.	To standardize the timeline globally and allow the algorithm to learn annual seasonality and weekly consumer purchasing behaviors.	Standardized temporal numerical inputs compatible with ML algorithms, preventing format errors during model training.
Recipe Unit Standardization: Converted all menu ingredients from volumetric units (spoons, cups) into standard weight (kilograms).	The raw market prices are strictly forecasted in THB/kg. Recipe proportions must match this unit for accurate total cost calculation.	A unified matrix proportion ready for direct multiplication with the forecasted prices to output accurate menu cost rankings.
Contextual Feature Engineering: Created an Important_Days binary flag (1 = Holiday/Festival, 0 = Normal Day).	Cultural events (e.g., Vegetarian Festival, Chinese New Year) trigger sudden, non-linear demand spikes that cannot be predicted by weather or fuel alone.	A crucial categorical feature that explicitly enables the models to anticipate and track episodic market shocks.

3.4 Model Selection, Implementation, and Hyperparameter Tuning To establish a highly accurate forecasting system capable of predicting ingredient prices with a 7-day horizon, three distinct machine learning architectures were comparatively evaluated. The rationale for selecting each model and their specific parameter customizations are summarized in Table 3.2

Table 3.2. Model Selection, Rationale, and Customization

Model	Architecture / Key Hyperparameters	Optimization & Training
Random Forest	n_estimators: 100 random_state: 42	No scaling required. Executed with parallel processing (n_jobs=-1).
SimpleRNN	Layer 1: LSTM (64 units, ReLU) Layer 2: Dropout (20%) Layer 3: Dense (32 units, ReLU) Output: Dense (1 unit)	Scaler: MinMaxScaler (X and y) Optimizer: Adam Loss: MSE Epochs: 50 Batch Size: 32
LSTM	Layer 1: LSTM (64 units, ReLU) Layer 2: Dropout (20%) Layer 3: Dense (32 units, ReLU) Output: Dense (1 unit)	Scaler: MinMaxScaler (X and y) Optimizer: Adam Loss: MSE Epochs: 50 Batch Size: 32

Walk-Forward Validation Strategy: Standard train-test splits are fundamentally flawed for time-series data. Therefore, a realistic "Walk-Forward Validation" approach was implemented. The models were initially trained on historical data. They then predicted prices for a 7-day window. Subsequently, the actual prices from that window were incorporated back into the training set, and the models were retrained. This dynamic loop meticulously simulates real-world deployment where forecasting systems continuously ingest new market data weekly.

3.5 End-to-End Data Pipeline for Food Price Forecasting

The diagram illustrates an end-to-end data pipeline (**see Appendix A, Figure A.7**), starting from four raw data sources that undergo data cleaning and feature engineering, such as converting the Buddhist Era (BE) to the Common Era (CE), handling missing values, and merging datasets. Ultimately, this pipeline systematically analyzes correlations within the processed data and utilizes the trained predictive models to forecast ingredient prices, seamlessly feeding into the cost-optimized meal recommendation system.

CHAPTER 4

RESULTS

This chapter presents the outcomes of the predictive modeling phase, focusing exclusively on the performance of the machine learning algorithms. It details the quantitative evaluation of the forecasting models and provides an in-depth discussion on the implications of the results, specifically regarding algorithm selection and feature importance.

4.1 Model Evaluation Results

The forecasting models were quantitatively evaluated using a 7-Day Walk-Forward Validation strategy on the final 20% of the dataset. The primary objective was to predict the continuous price of fresh ingredients (using Coriander as the benchmark). Performance was measured using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Table 4.1 Performance Comparison of Forecasting Models (Coriander Price)

Model	MAE(THB/kg)	RMSE(THB/kg)	MAPE (%)
Random Forest	19.5457	28.4755	17.22%
SimpleRNN	25.3117	35.6134	22.65%
LSTM	23.3542	33.9325	20.73%

Evaluation: According to the results, the model achieved a Mean Absolute Error (MAE) of approximately 19.65 THB/kg. This means that when the Random Forest predicts coriander prices, its predictions typically deviate from the actual price by about ± 20 THB on average.

To further understand the mechanics driving these results, Table 4.2 outlines the best hyperparameters utilized for each model, alongside the top three most influential input features that dictated their predictions.

Table 4.2. Best Parameters and Top 3 Important Features per Model

Model	Best Parameters	Top 3 Input Features
Random Forest	Best Parameters: {'max_depth': 20, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}	Price_chicken, Price_spring_onion, Price_peanut
SimpleRNN	units: 64, dropout: 0.2, epochs: 50	Price_shallot, Price_onion, Price_tamarind_juice
LSTM	units: 64, dropout: 0.2, epochs: 50	Price_tamarind_juice, Avg_Temperature_C, Avg_Soil_Moisture

This chapter details the outcomes of the end-to-end analytical pipeline. It transitions from exploratory data insights and feature engineering outcomes to the comparative evaluation of the forecasting algorithms, culminating in the primary technical contribution of this study.

4.2 Main Contribution: Cost-Optimized Menu Recommendation System

By validating the Random Forest as a robust predictive engine, this study achieves its primary technical contribution. Rather than isolating the output as merely theoretical price forecasts, the predicted values (THB/kg) are systematically matrix-multiplied with standard recipe proportions to calculate the future cumulative cost of 5 popular Thai dishes. A preliminary mock-up (**see Appendix G, Figures G.1 and G.2**) demonstrates how the system dynamically ranks meal costs. This allows the system to proactively recommend meals, empowering users to avoid actively surging ingredients and transforming raw machine learning outputs into an actionable, cost-saving tool for everyday consumers.

CHAPTER 5

CONCLUSION AND FUTURE WORK

CHAPTER 5 DISCUSSION, CONCLUSION, AND FUTURE WORK

5.1 Discussion and Technical Insights

Before concluding the study, it is crucial to analyze the findings from the model evaluation. The results yielded several critical insights into analyzing highly volatile time-series agricultural data:

- **Why did Random Forest outperform Deep Learning?:** Contrary to the initial assumption that advanced sequential models (LSTM) would yield the highest accuracy, the Random Forest Regressor emerged as the superior model. Visual inspection of the Walk-Forward Validation plots (see Appendix E, Figure E.5) provides justification. Deep models like LSTM are designed to identify long-term, continuous trends; thus, they tend to "smooth out" predictions, conservatively underestimating the magnitude of sudden price spikes. In contrast, the Random Forest's tree-based architecture is highly adept at learning non-linear thresholds and strict conditional rules. It effectively captured abrupt, episodic market shocks without the high variance observed in the SimpleRNN.
- **Interpretation of Feature Importance:** The extraction of feature importance (refer to Table 4.2) mathematically reveals distinct learning behaviors across the models. For the Random Forest and SimpleRNN, contemporaneous ingredient prices (e.g., Price_chicken, Price_shallot, Price_tamarind_juice) emerged as the dominant features. This implies that the target price is heavily influenced by the broader market movements of other key ingredients, reflecting shared supply chain dynamics. Interestingly, the LSTM model placed higher importance on continuous environmental variables (Avg_Temperature_C and Avg_Soil_Moisture). This demonstrates that while tree-based models excel at capturing immediate cross-commodity market shocks, deep sequential models prioritize long-term climatic trends to predict price shifts.

5.2 Conclusion

This study successfully developed an end-to-end machine learning pipeline that addresses the volatility of agricultural commodity prices by translating raw point forecasts into a practical "Cost-Optimized Menu Recommendation System." By constructing a comprehensive 5-year multivariate dataset and resolving structural missingness via Forward Fill, the study established a robust modeling foundation. Evaluated through a 7-Day Walk-Forward Validation strategy, the Random Forest Regressor emerged as the optimal engine, outperforming deep learning models (RNN, LSTM) by effectively capturing non-linear market shocks. Ultimately, by systematically matrix-multiplying the forecasted ingredient prices with standard recipe proportions, the system dynamically ranks meal costs. This paradigm shift from isolated price tracking to actionable utility empowers consumers and food vendors to proactively mitigate market shocks and strategically optimize their weekly budgets.

5.3 Recommendations for Practical Application

- **For Consumers and Food Vendors:** The system serves as a strategic weekly budget planner. During periods of anticipated high volatility, users should substitute highly sensitive menus with the algorithm's recommended cost-effective alternatives.
- **For Data Engineers:** Prioritizing feature engineering over complex model architectures is highly recommended. Explicitly modeling macroeconomic anchors (e.g., diesel prices) and calendar flags yields higher predictive stability than processing unscaled weather data through deep neural networks.

5.4 Future Work

- **Addressing Incomplete Ingredient Coverage:** Currently, the model predicts prices for 13 key ingredients. Complex dishes requiring numerous unlisted ingredients may not reflect their true total cost. Future iterations must expand the data collection pipeline and feature space to track a wider array of minor vegetables and condiments.
- **Integration of Personalized User Preferences:** Future enhancements should incorporate customized user profiles, factoring in dietary restrictions and flavor preferences via consumer surveys, thereby bridging quantitative cost forecasting with qualitative user needs.
- **Full System Deployment:** Transitioning the current static mock-up into a fully deployed, live web or mobile application will facilitate seamless, interactive daily meal planning for end-users.

REFERENCES

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APPENDIX A

Data Information

```
[7]: chicken_df = pd.read_csv('chicken.csv')
display(chicken_df.head())
display(chicken_df.tail())
```

	วันที่	ประเภท	สินค้า	หน่วย	ราคา ต่ำสุด	ราคา สูงสุด	ราคาเฉลี่ย
0	4/1/2564	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	60	65	62.5
1	5/1/2564	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	60	65	62.5
2	6/1/2564	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	60	65	62.5
3	7/1/2564	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	60	65	62.5
4	8/1/2564	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	60	65	62.5

	วันที่	ประเภท	สินค้า	หน่วย	ราคา ต่ำสุด	ราคา สูงสุด	ราคาเฉลี่ย
1288	24/12/2568	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	80	90	85.0
1289	25/12/2568	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	80	90	85.0
1290	26/12/2568	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	80	90	85.0
1291	29/12/2568	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	80	90	85.0
1292	30/12/2568	ขายปลีก	ไก่สดชำแหละ น่อง สะโพก	บาท/กก.	80	90	85.0

```
chicken_df = pd.read_csv('chicken.csv')
chicken_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1293 entries, 0 to 1292
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   วันที่           1293 non-null  object
1   ประเภท         1293 non-null  object
2   สินค้า         1293 non-null  object
3   หน่วย          1293 non-null  object
4   ราคา ต่ำสุด    1293 non-null  int64
5   ราคา สูงสุด    1293 non-null  int64
6   ราคาเฉลี่ย     1293 non-null  float64
dtypes: float64(1), int64(2), object(4)
memory usage: 70.8+ KB
```

Figure A.1 Information of ingredient price (Chicken)

```
[8]: oil_2565_df = pd.read_csv('oil_2565.csv')
display(oil_2565_df.head())
display(oil_2565_df.tail())
```

	วันที่	บาท/ลิตร	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
0	NaN	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S	ไอโซพรีน 5 ดีเซล S
1	22/12/2565	43.66	34.94	42.54	32.69	32.54	34.18	34.45
2	21/12/2565	43.66	34.94	42.84	32.69	32.54	34.18	34.45
3	16/12/2565	43.66	34.94	42.84	32.99	32.84	34.48	34.75
4	15/12/2565	43.66	34.94	-	32.69	32.54	34.18	34.45

	วันที่	บาท/ลิตร	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
97	20/01/2565	35.96	29.94	-	24.94	31.64	32.88	33.15
98	14/01/2565	35.86	29.84	-	24.64	31.14	32.38	32.65
99	11/1/2565	35.86	29.84	-	24.34	30.64	31.88	32.15
100	8/1/2565	35.46	29.44	-	24.14	30.24	31.48	31.75
101	5/1/2565	35.06	29.04	-	24.14	30.24	31.48	31.75

```
oil_2565 = pd.read_csv('oil_2565.csv')
oil_2565.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102 entries, 0 to 101
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   วันที่      101 non-null   object
1   บาท/ลิตร    102 non-null   object
2   Unnamed: 2  102 non-null   object
3   Unnamed: 3  102 non-null   object
4   Unnamed: 4  102 non-null   object
5   Unnamed: 5  102 non-null   object
6   Unnamed: 6  102 non-null   object
7   Unnamed: 7  102 non-null   object
dtypes: object(8)
memory usage: 6.5+ KB
```

Figure A.2 Information of Diesel

```
[10]: thai_calendar_2021_english_df = pd.read_csv('thai_calendar_2021_english.csv')
display(thai_calendar_2021_english_df.head())
display(thai_calendar_2021_english_df.tail())
```

	Date	Day_Name	Month_Name	Year	Important_Events	Day_Type
0	1/1/2021	Friday	January	2021	New Year's Day	Yes
1	2/1/2021	Saturday	January	2021	NaN	Weekend
2	3/1/2021	Sunday	January	2021	NaN	Weekend
3	4/1/2021	Monday	January	2021	NaN	No
4	5/1/2021	Tuesday	January	2021	NaN	No

	Date	Day_Name	Month_Name	Year	Important_Events	Day_Type
360	27/12/2021	Monday	December	2021	NaN	No
361	28/12/2021	Tuesday	December	2021	King Taksin Memorial Day	Yes
362	29/12/2021	Wednesday	December	2021	NaN	No
363	30/12/2021	Thursday	December	2021	NaN	No
364	31/12/2021	Friday	December	2021	New Year's Eve	Yes

```
: thai_calendar_2021_english = pd.read_csv('thai_calendar_2021_english.csv')
thai_calendar_2021_english.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 365 entries, 0 to 364
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Date             365 non-null   object
1   Day_Name         365 non-null   object
2   Month_Name       365 non-null   object
3   Year             365 non-null   int64
4   Important_Events 101 non-null   object
5   Day_Type         365 non-null   object
dtypes: int64(1), object(5)
memory usage: 17.2+ KB
```

Figure A.3 Information of Calendar

```
Recipes_df = pd.read_csv('Thai_Popular_Recipes_20_Menus_EN.csv')
display(Recipes_df.head())
display(Recipes_df.tail())
```

	Menu	Ingredient	Quantity	Unit
0	1. Pad Kra Pao Moo Sap (Holy Basil Stir-Fry)	Minced Pork	300	Grams
1	1. Pad Kra Pao Moo Sap (Holy Basil Stir-Fry)	Holy Basil Leaves	1	Cup
2	1. Pad Kra Pao Moo Sap (Holy Basil Stir-Fry)	Minced Garlic	1	Tbsp
3	1. Pad Kra Pao Moo Sap (Holy Basil Stir-Fry)	Red Bird's Eye Chili (Pounded)	5-10	Pod
4	1. Pad Kra Pao Moo Sap (Holy Basil Stir-Fry)	Oyster Sauce	1.5	Tbsp

	Menu	Ingredient	Quantity	Unit
165	20. Kai Jeow Moo Sap	Eggs	3	Eggs
166	20. Kai Jeow Moo Sap	Minced Pork	50	Grams
167	20. Kai Jeow Moo Sap	Fish Sauce/Soy Sauce	1	Tsp
168	20. Kai Jeow Moo Sap	Spring Onion (Sliced)	1	Stalk
169	20. Kai Jeow Moo Sap	Vegetable Oil (For Frying)	0.5	Cup

Figure A.4 Information of Recipes

```

[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[5]: weather_df = pd.read_csv('weather_4provinces_daily_2021_2025.csv')
display(weather_df.head())
display(weather_df.tail())

```

	Date	Province	Role	Temperature_C	Rainfall_mm	Humidity_Percent	Soil_Moisture
0	1/1/2021	PathumThani_Hub	Market_Consumer	22.08	0.0	52.29	0.29
1	2/1/2021	PathumThani_Hub	Market_Consumer	22.73	0.0	52.92	0.29
2	3/1/2021	PathumThani_Hub	Market_Consumer	24.00	0.0	56.33	0.29
3	4/1/2021	PathumThani_Hub	Market_Consumer	25.47	0.0	58.04	0.29
4	5/1/2021	PathumThani_Hub	Market_Consumer	26.65	0.0	58.83	0.29

	Date	Province	Role	Temperature_C	Rainfall_mm	Humidity_Percent	Soil_Moisture
7299	27/12/2025	NakhonPathom_Meat	Meat_Source	26.25	0.0	66.33	0.16
7300	28/12/2025	NakhonPathom_Meat	Meat_Source	26.17	0.0	64.50	0.15
7301	29/12/2025	NakhonPathom_Meat	Meat_Source	25.64	0.0	66.46	0.16
7302	30/12/2025	NakhonPathom_Meat	Meat_Source	25.16	0.0	69.96	0.16
7303	31/12/2025	NakhonPathom_Meat	Meat_Source	24.75	0.0	72.04	0.16

```

[110]: weather_df = pd.read_csv('weather_4provinces_daily_2021_2025.csv')
weather_df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7304 entries, 0 to 7303
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  7304 non-null  object
1   Province              7304 non-null  object
2   Role                  7304 non-null  object
3   Temperature_C         7304 non-null  float64
4   Rainfall_mm           7304 non-null  float64
5   Humidity_Percent      7304 non-null  float64
6   Soil_Moisture          7304 non-null  float64
dtypes: float64(4), object(3)
memory usage: 399.6+ KB

```

Figure A.5 Information of Weather (Pathumthani)

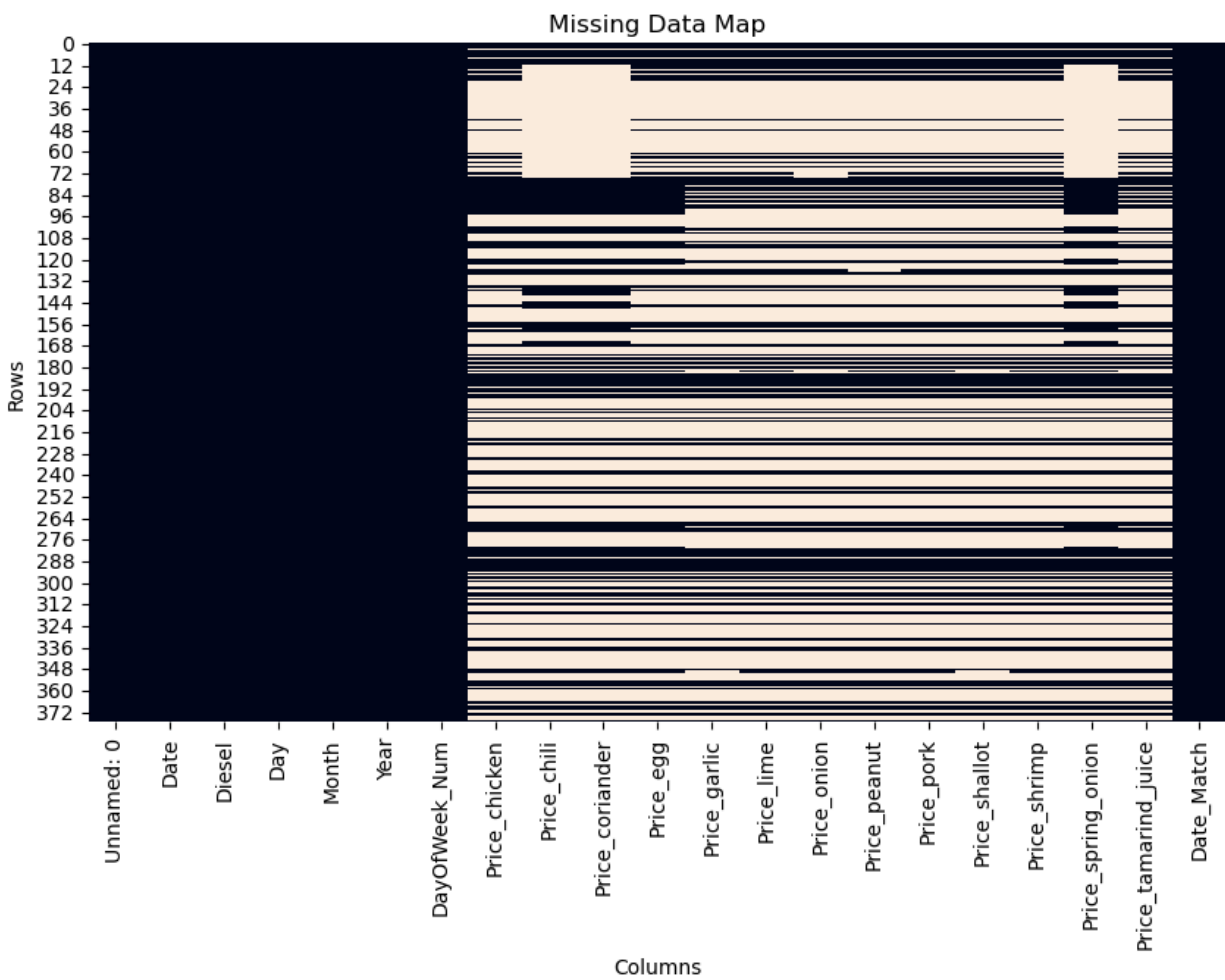


Figure A.6: Overview of Missing Feature Data

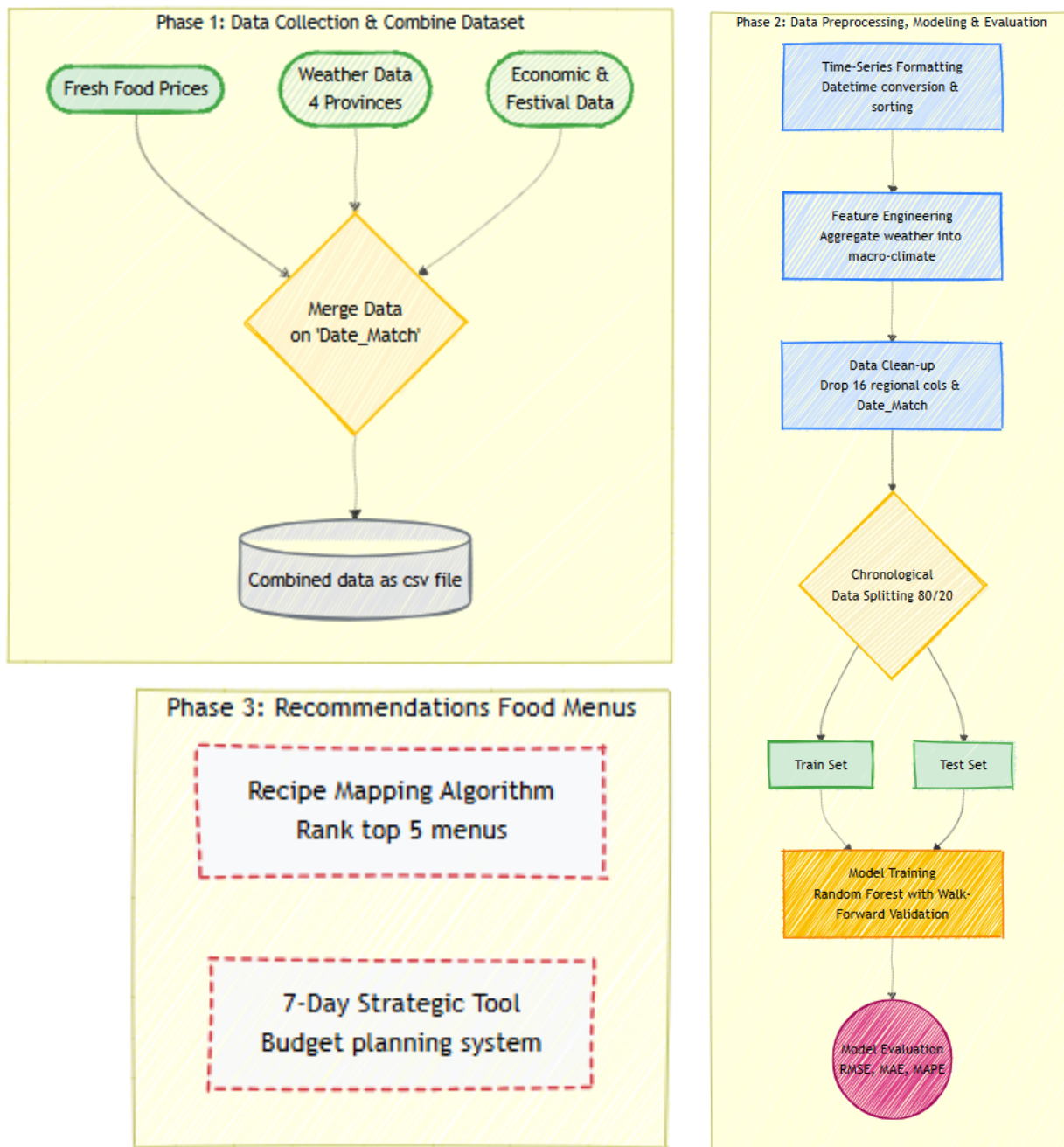


Figure A.7: Overall Architecture Diagram

Appendix B

Dataset Summary

Table B.1. Summary of Collected Datasets

Dataset	Type	Key Features	<i>Instances/Rows</i>	<i>Missing Values</i>
Weather	Numerial	Temp, Rainfall, Humidity, Soil Moisture	7,304	Non-null
Agriculture Price (Target)	String/Numeria l	Avg Retail Prices (13 ingredients)	1,293 (Business days)	Non-null
Diesel Price	String/Numeria l	Hi Diesel S price (Daily)	.~102 day / year	NaN on Row 0
Calendar	Numerial	Data compatible with ML	1,826	264 NaN
Recipe	String	Ingredients & Quantities for 20 menus	170	Non-null

APPENDIX C

Summary of Feature Engineering

Table C.1. Summary of Feature Engineering Techniques

Dataset	Type	Key Features
Contextual Feature Engineering	Important_Days (Binary Flag)	Convert holiday or festival names (e.g., Songkran, New Year, Vegetarian Festival) into numerical values: 0 (normal day) and 1 (important day) to capture their impact on demand and meat prices.
	Macroeconomic Indicator	Diesel fuel prices are used as an independent variable (X) to reflect the transportation and logistics costs of agricultural products.
Temporal Feature Extraction	Decomposing Dates	The date format (DD/MM/YYYY) is decomposed into Day, Month, and Year so that the model can capture seasonal patterns (e.g., lime prices tend to increase during the dry season).
	DayOfWeek_Num	A variable indicating the day of the week is created (0 = Monday, 6 = Sunday) to capture different purchasing behaviors between weekdays and weekends.
	Date Standardization	Convert the Buddhist Era (Thai Solar Calendar) to the Gregorian calendar by subtracting 543 to ensure international standardization and accurate integration of data from multiple sources.
Weather Data Reshaping	Pivoting (Long to Wide)	Weather data is transformed from a “long format” (data from each province arranged vertically) into a “wide format” (one row per day) so that all variables from all provinces appear in the same row.
	Column Flattening	New variable names are created to specify both the province and the data type (e.g., PathumThani_Hub_Temperature_C) to reduce the complexity of MultiIndex data structures.
Lag Feature Creation	Price Momentum	Historical price data (e.g., price at day t-1 or t-7) is used as independent variables to help forecast prices 7 days ahead (7-Day Forecast Horizon) based on past price trends.
Missing Value Handling (Imputation)	Forward Fill (ffill)	For agricultural product and fuel prices not updated on weekends, forward filling is applied by assuming that prices on non-trading days remain the same as the most recently announced price (e.g., Friday’s price).
Recipe Unit Standardization	Conversion to Kilograms	Ingredient quantities in recipes are converted from various units (e.g., cups, tablespoons, grams) into kilograms (kg) so that costs can be directly calculated using market prices reported in “Baht per kilogram.”

APPENDIX D

Data Dictionary

Table D.1. Consolidated Data Dictionary

Category	Feature Name	Data Type	Description
Temporal Identifiers	Date_Match	datetime64	Standardized daily timeline acting as the primary key (Gregorian).
	Day, Month, Year	int64	Extracted numerical values for day, month, and year.
	Decomposing Dates	int64	Numerical day of the week (0 = Monday, 6 = Sunday) for weekly trends.
Meteorological Variables (Measured across 4 provinces)	Temperature_C	float64	Average daily regional temperature (°C).
	Rainfall_mm	float64	Total daily regional precipitation (mm).
	Humidity_Percent	float64	Average daily relative humidity (%).
	Soil_Moisture	float64	Regional soil moisture index/percentage.
Macroeconomic Indicator	Diesel	float64	Standard daily retail price of Diesel (THB/Liter) reflecting logistics costs.
Microeconomic Variables	Price_[Ingredient]	float64	Average daily market prices for 13 key ingredients (Chicken, Pork, Egg, Shrimp, Lime, Chili, Coriander, Garlic, Onion, Peanut, Shallot, Spring Onion, Tamarind Juice).
Contextual Feature	Important_Days	int64	Binary indicator capturing major holidays/festivals (1 = Holiday, 0 = Normal Day).

APPENDIX E

SUPPLEMENTARY VISUALIZATIONS AND EDA CHARTS

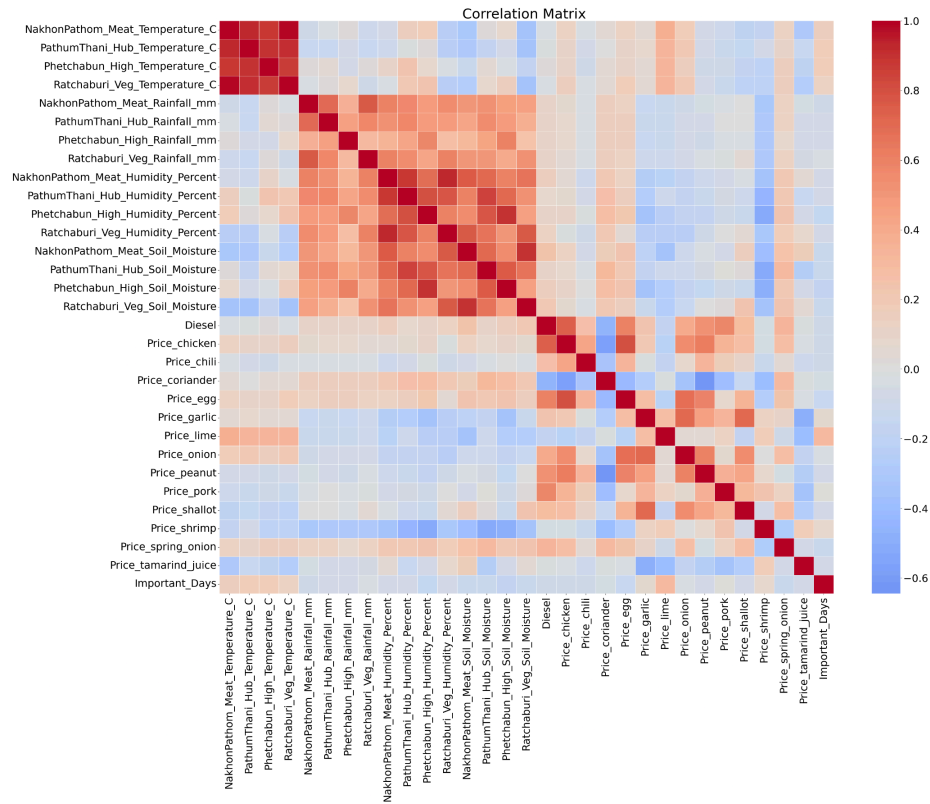


Figure E.1. Pearson Correlation Heatmap of All Features

The correlation matrix highlighting severe spatial multicollinearity among provincial weather variables (dark red clusters, $r > 0.8$) and the macroeconomic influence of diesel prices on livestock.

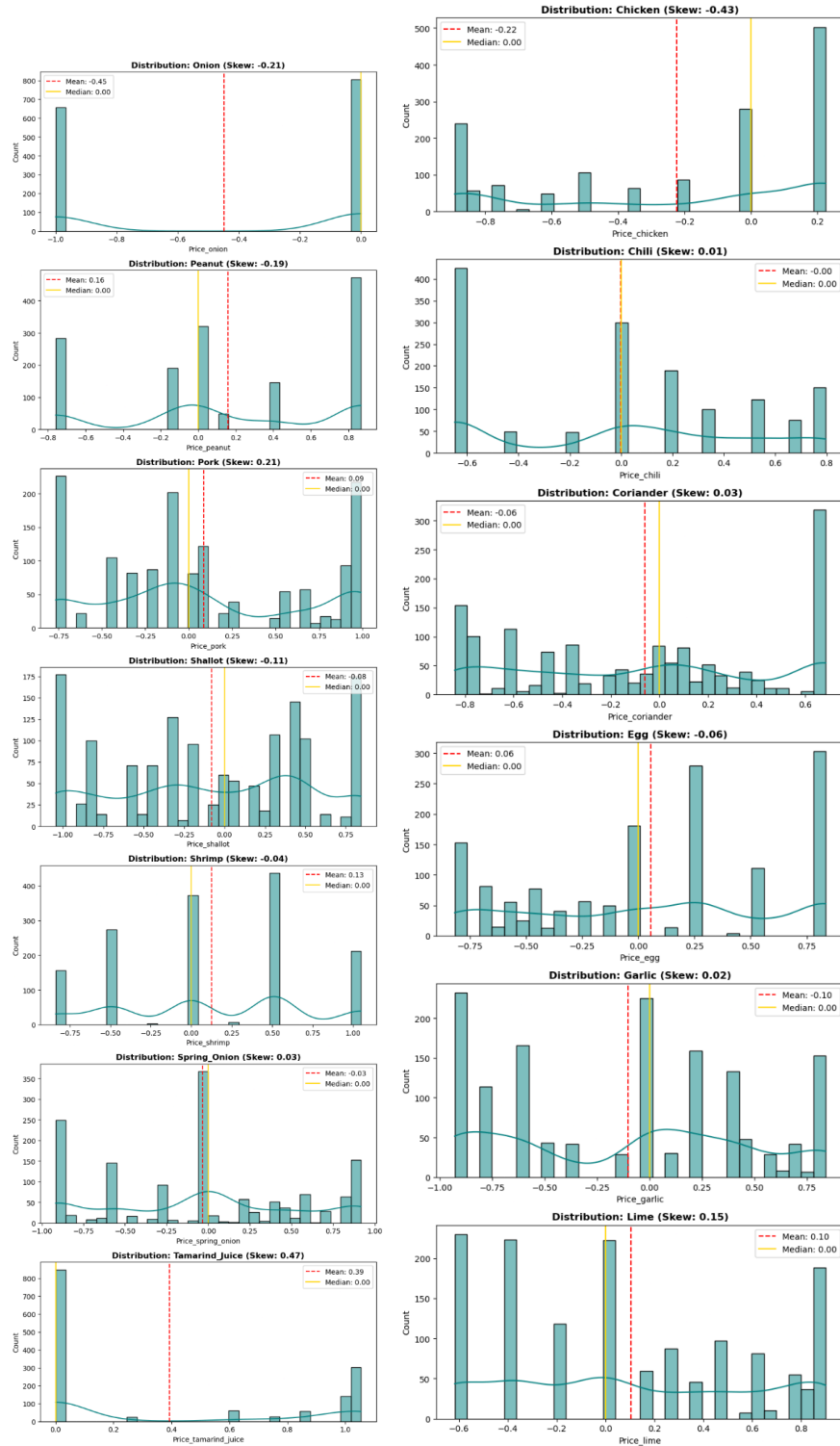


Figure E.2. Distribution and Skewness of Target Variables

Histograms displaying the data distribution of key ingredients. Note the severe right-skewness in highly volatile items like Chili and Lime, indicating frequent extreme price surges.

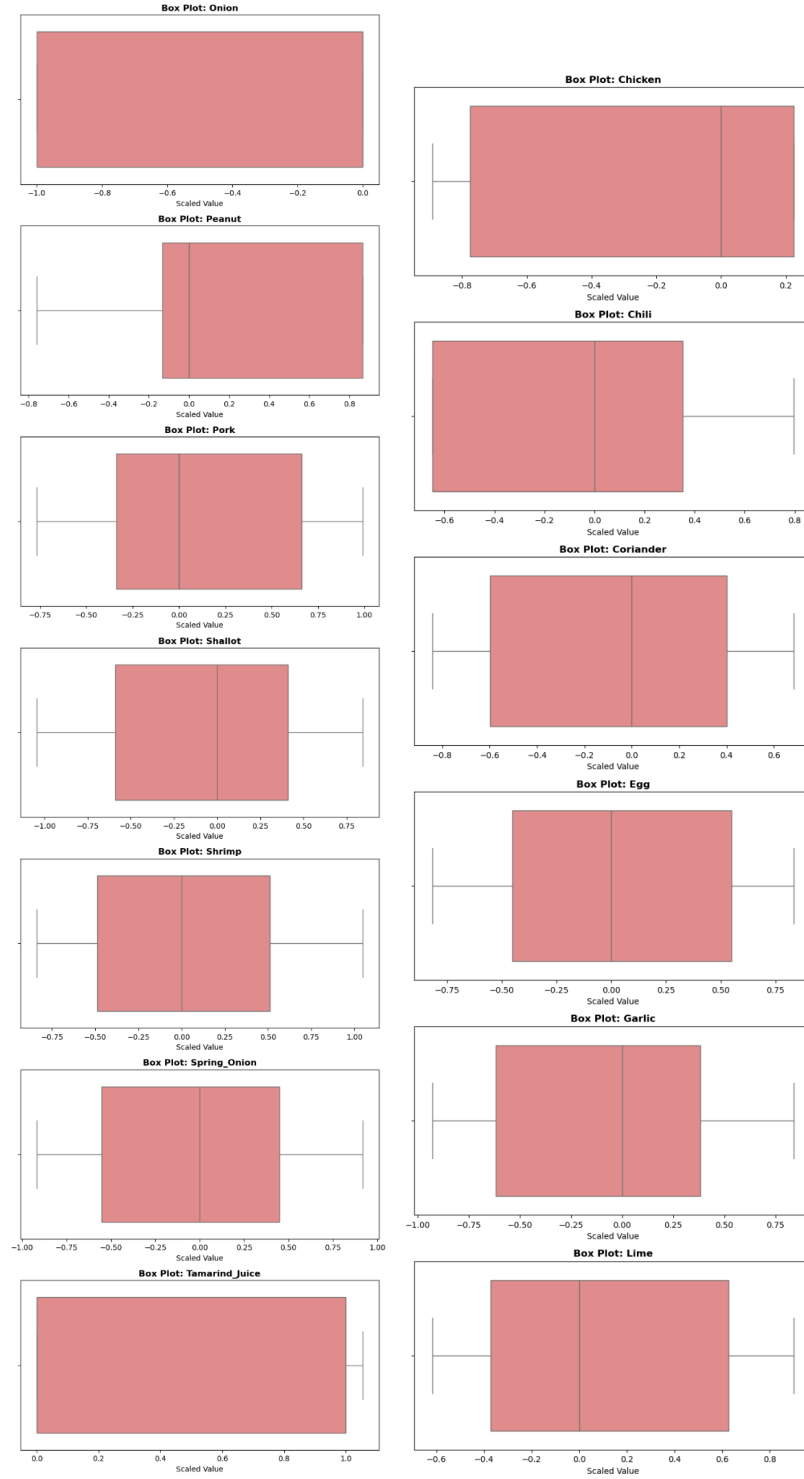


Figure E.3. Boxplot Analysis for Outlier Detection

Boxplots of the target variables confirming the presence of significant extreme upper outliers, particularly in vegetable and root crop categories during market shocks.

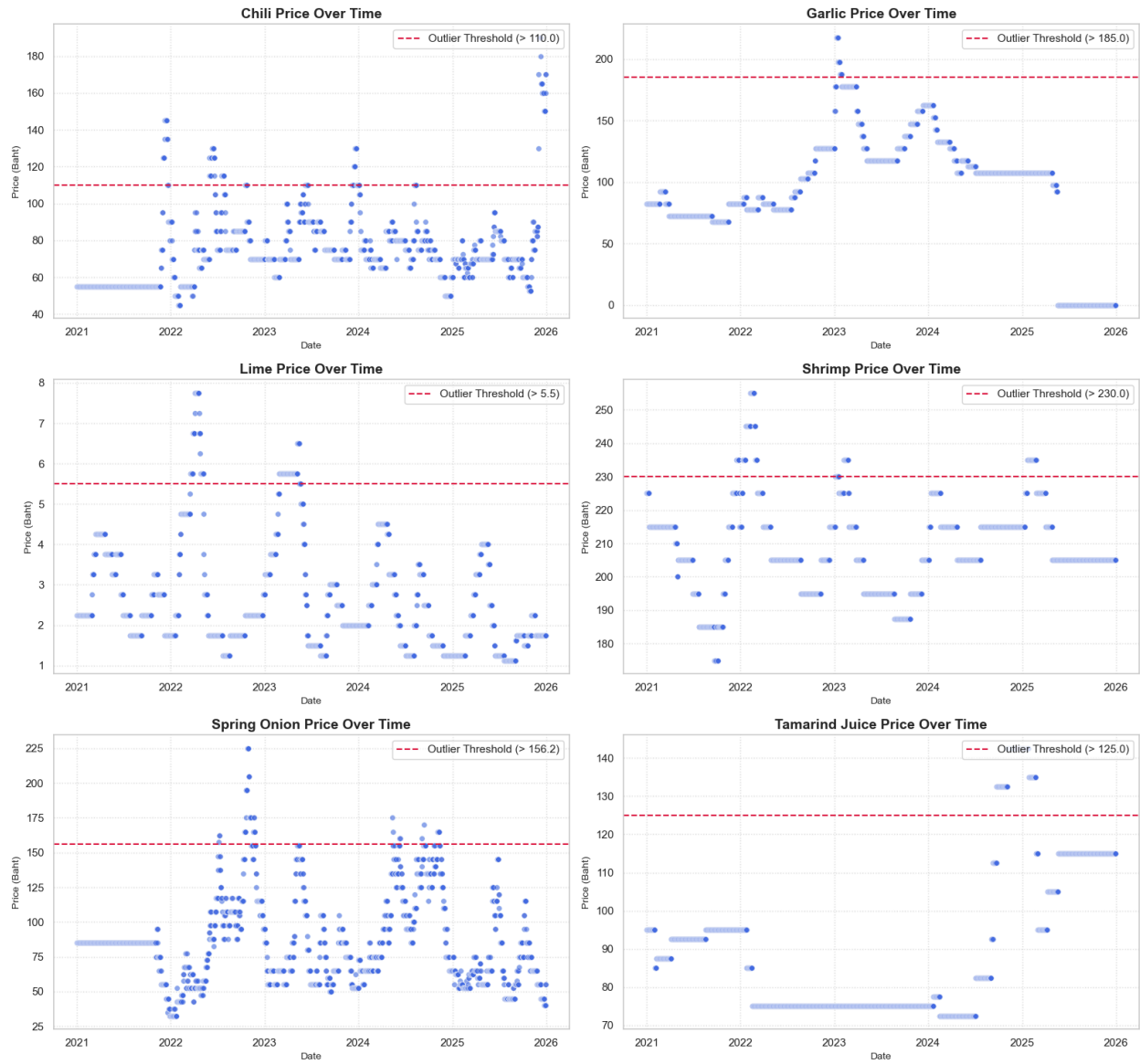


Figure E.4. Time-Series Volatility and Outlier Thresholds

Longitudinal time-series plots with predefined outlier thresholds (red dashed lines). The visuals confirm that price spikes are episodic and non-linear, often aligning with festivals or seasonal droughts.

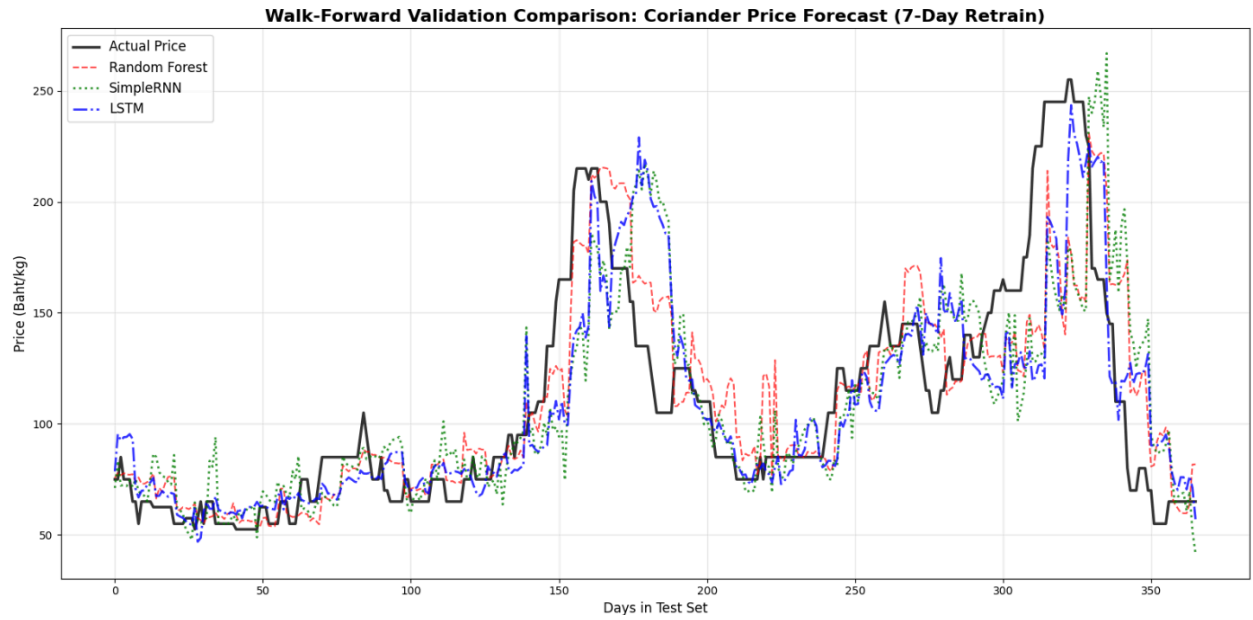


Figure E.5. Walk-Forward Validation Comparison (Actual vs. Predicted)

Walk-Forward Validation results comparing actual prices (black line) against the Random Forest, LSTM, and SimpleRNN models. The Random Forest demonstrates superior capability in tracking extreme price volatility and non-linear peaks.

APPENDIX F

SUPPLEMENTARY DATA TABLES

Table F.1. Summary of Key EDA Findings and Market Dynamics

Market Dynamic	Description & Insight
Severe Spatial Multicollinearity	Weather variables from all 4 provinces exhibit extremely high correlations ($r > 0.8$), indicating nearly identical climatic cycles and necessitating dimensionality reduction.
Macroeconomic Driver	Diesel fuel price ($r > 0.5$) acts as a primary macroeconomic anchor influencing livestock prices (chicken, eggs, pork), reflecting embedded logistics costs.
Vegetable Market Co-movements	Prices of root crops and vegetables (e.g., peanuts, onions, garlic) move synchronously, reflecting shared harvesting cycles and supply chain constraints.
Demand Shifts during Festivals	The Important_Days binary flag strongly correlates with price spikes in high-demand proteins (pork, chicken) during festivals.
Inverse Relationships	Notable negative correlations exist, such as shrimp prices being negatively sensitive to Phetchabun's humidity ($r < -0.5$), indicating precise environmental vulnerabilities.

APPENDIX G

Mock up menu price prediction

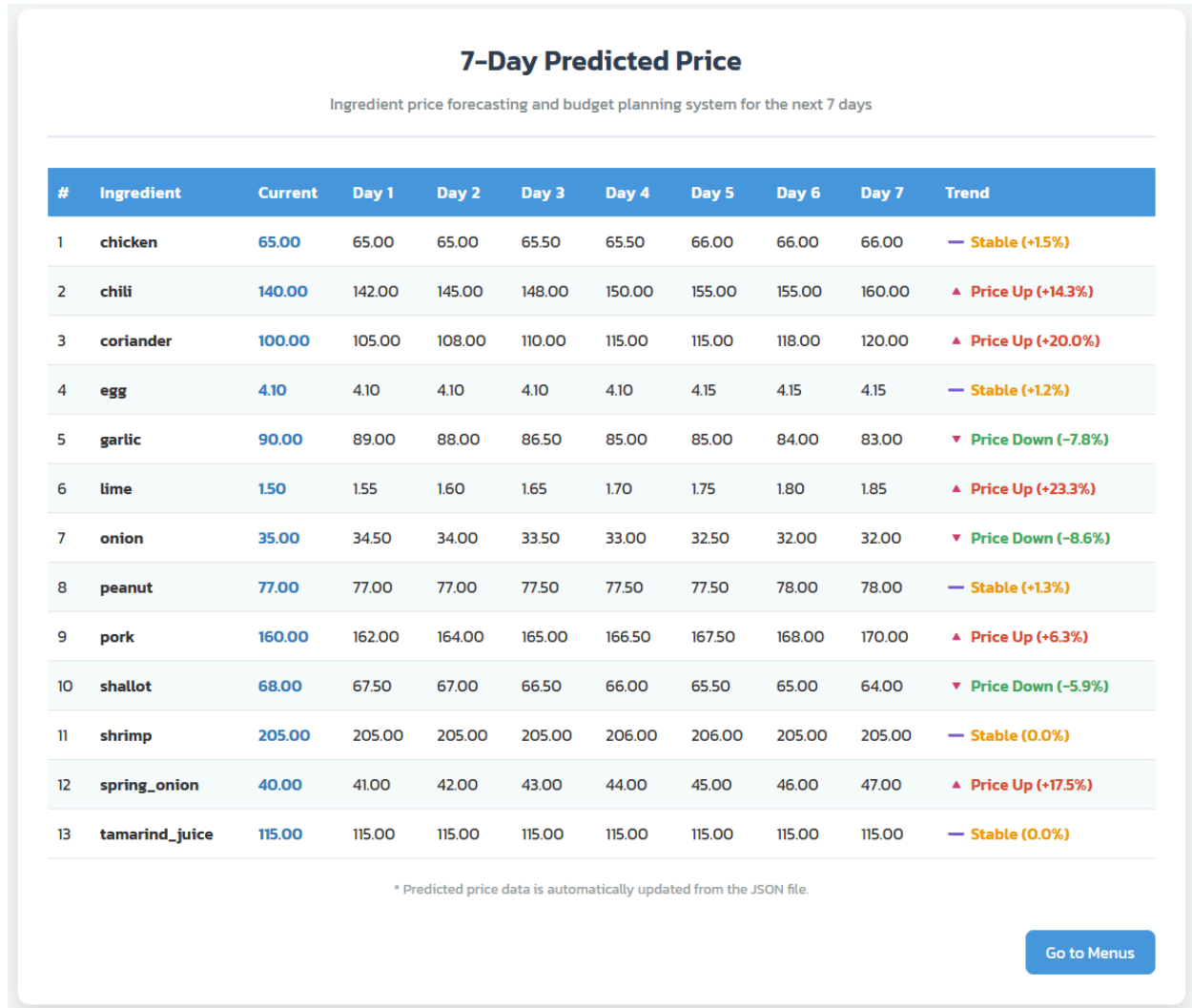


Figure G.1. 7-Day predicted Price

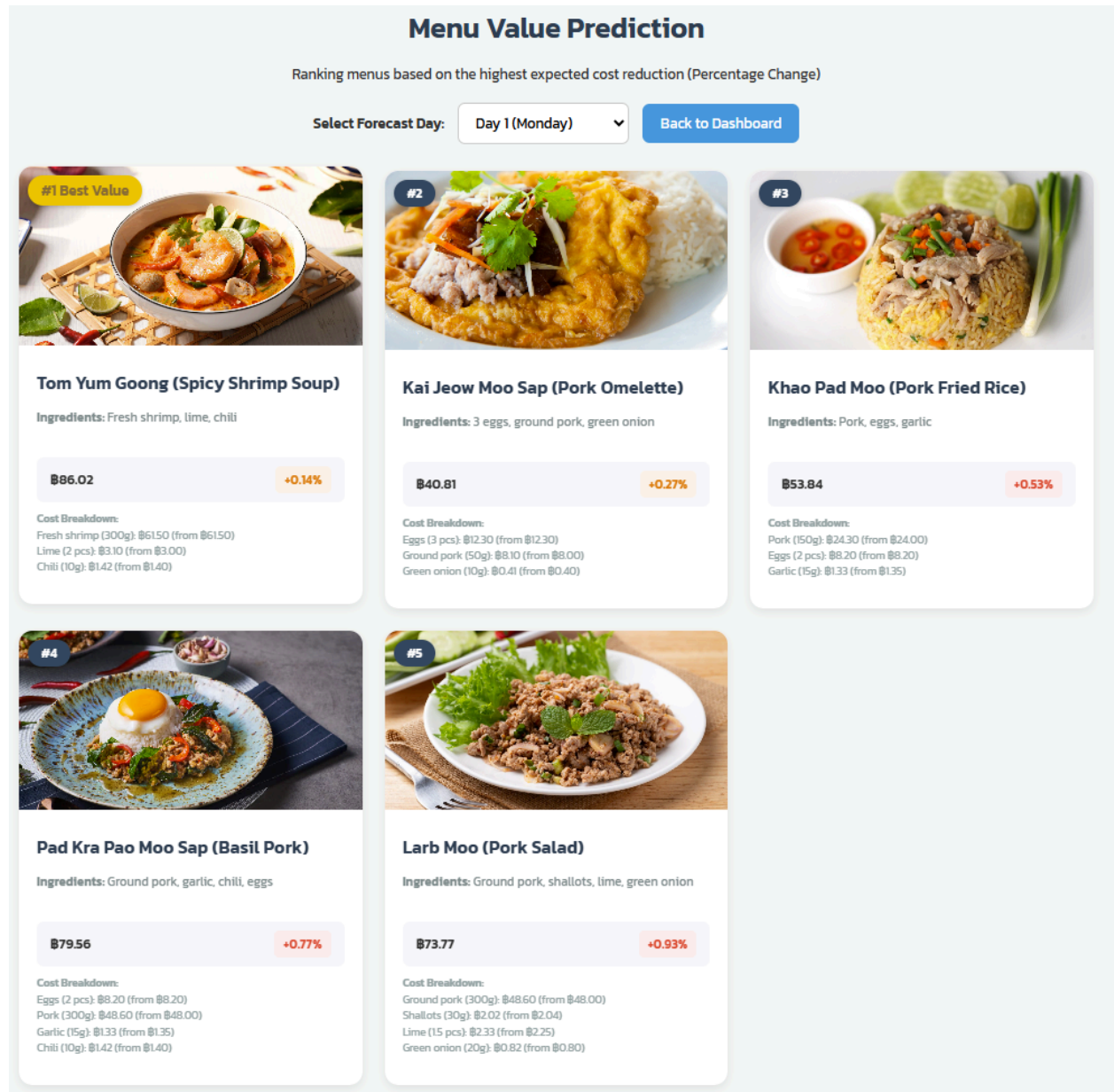


Figure G.2. Menu Price Prediction